

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: MemX-Rev-Ch.sil etc.

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For: Spruce

26 total sheets = 25 pages, 1 copy.

Problems encountered:

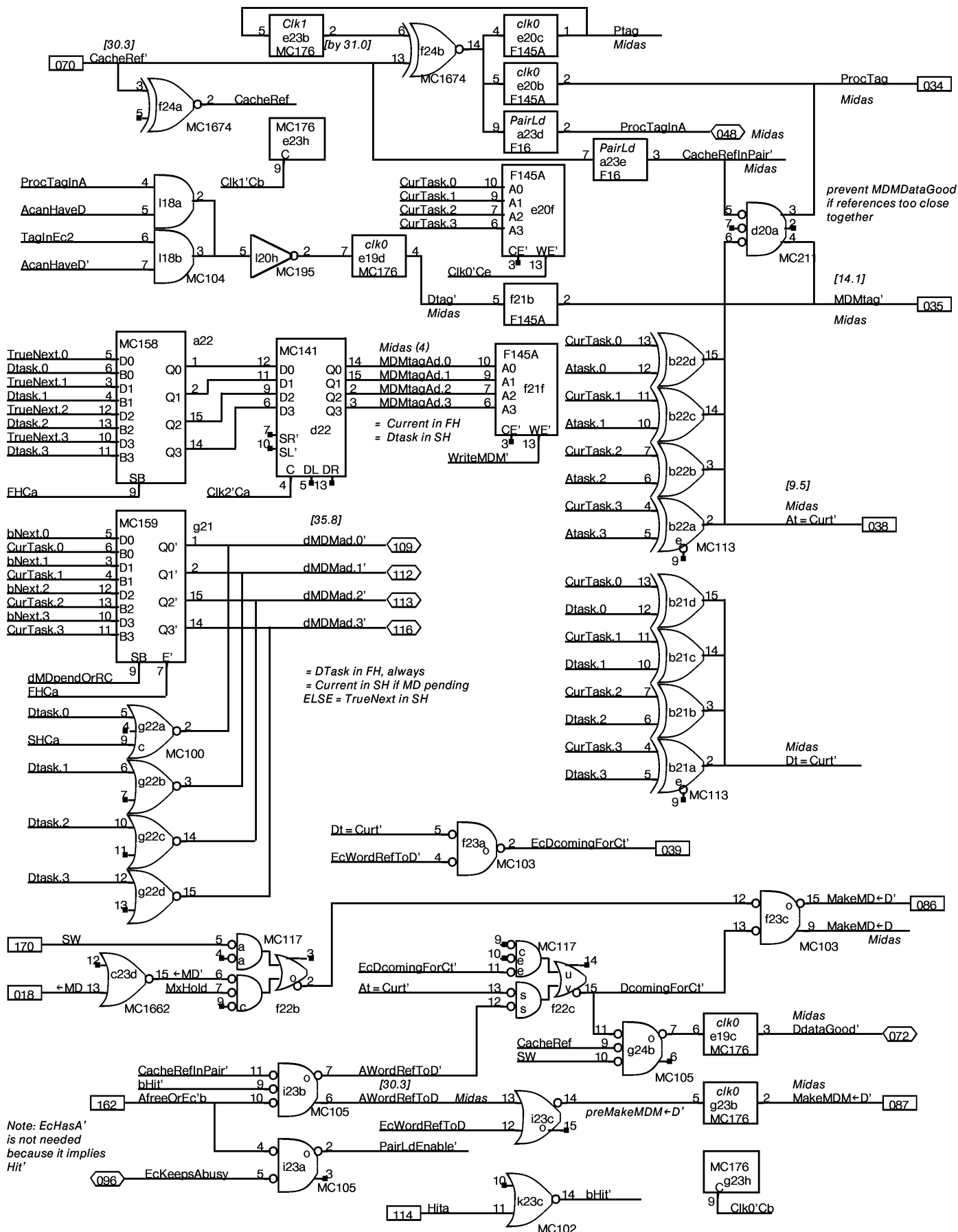
Font HELVETICA24B substituted for font HELVE24.

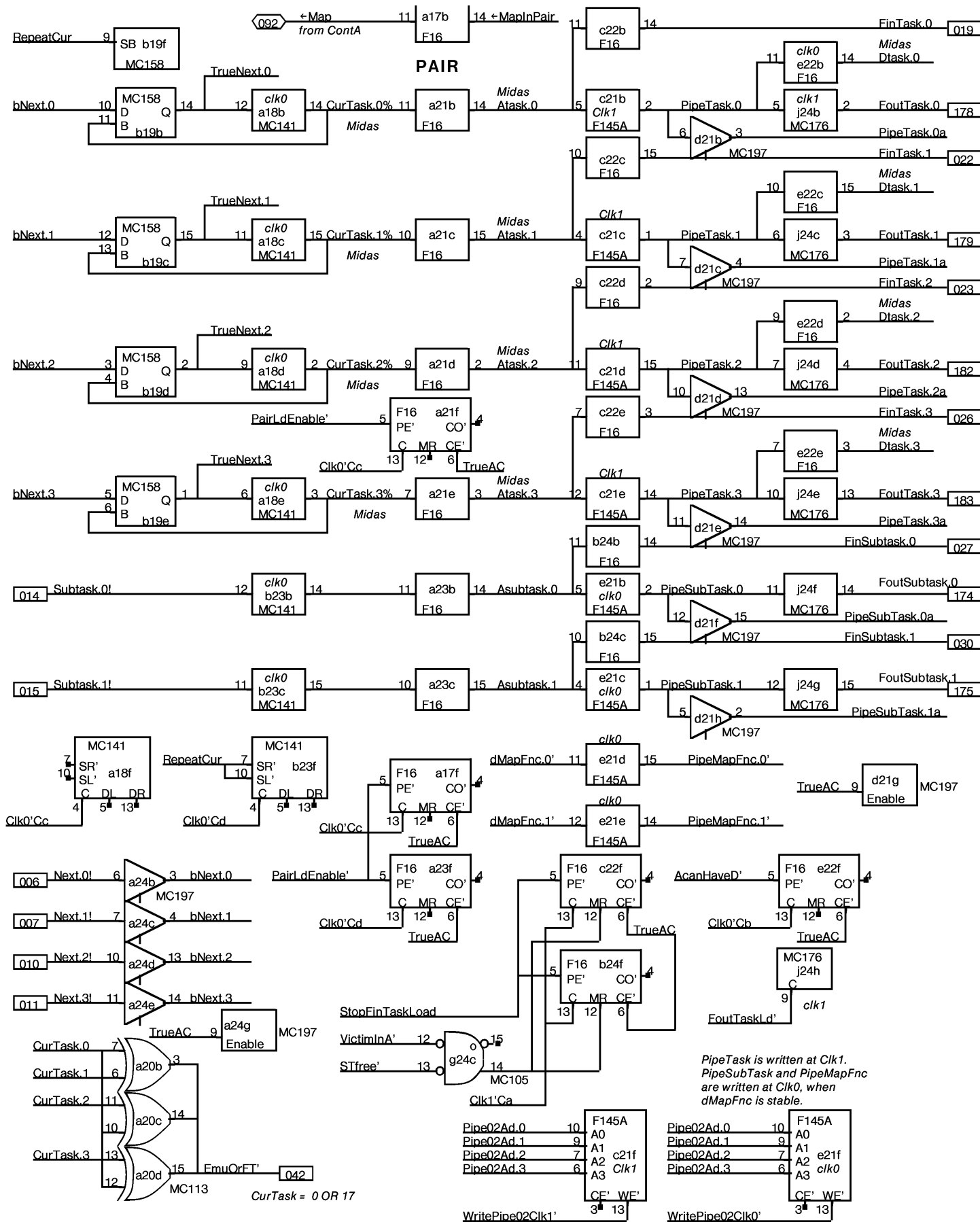
DORADO SCHEMATICS

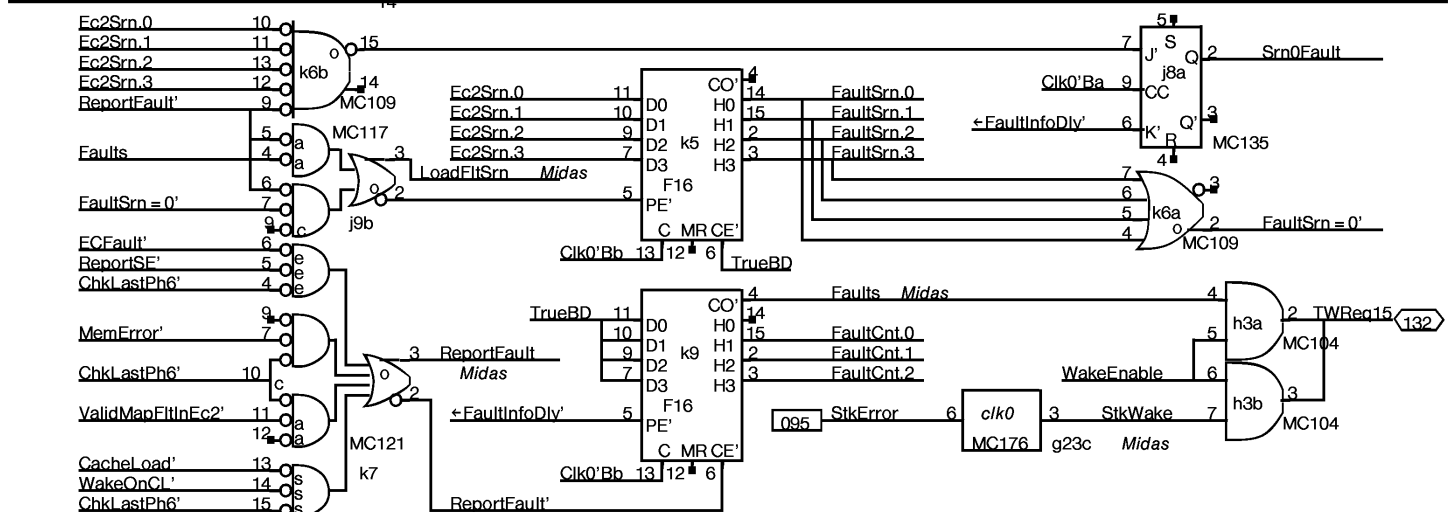
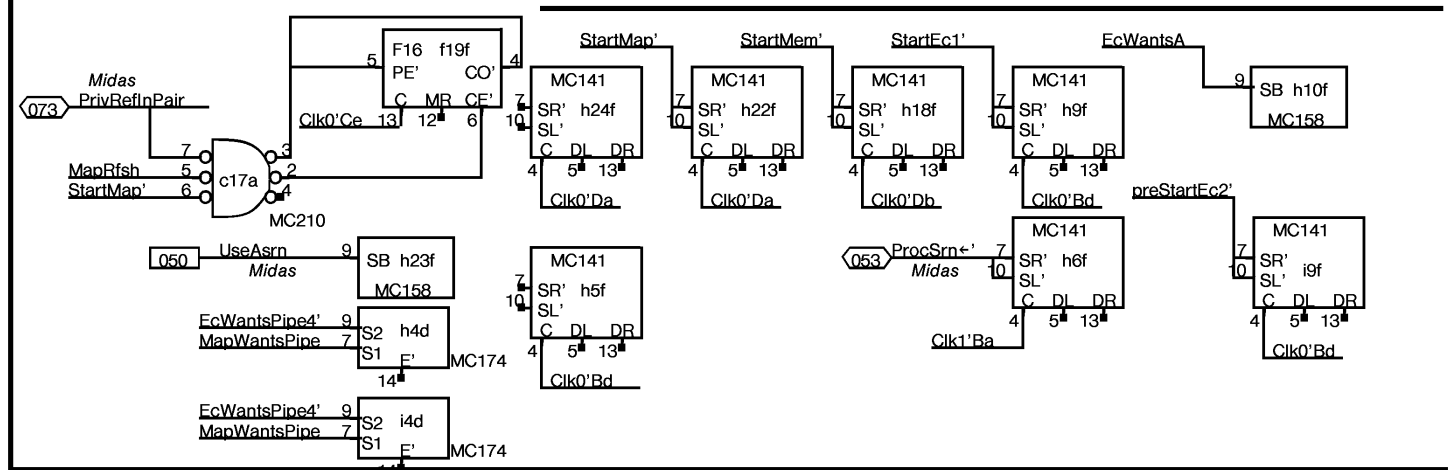
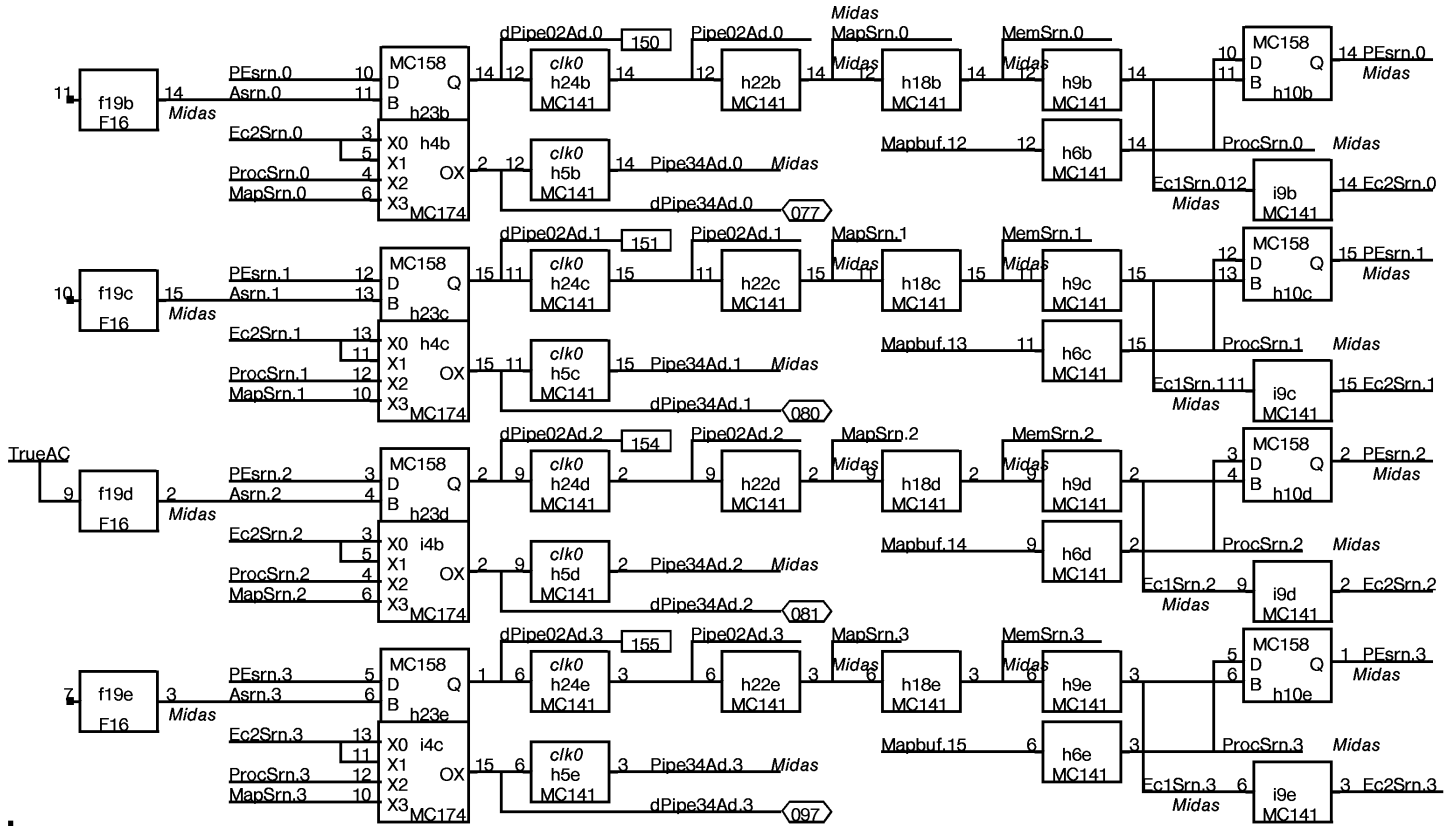
Mem X

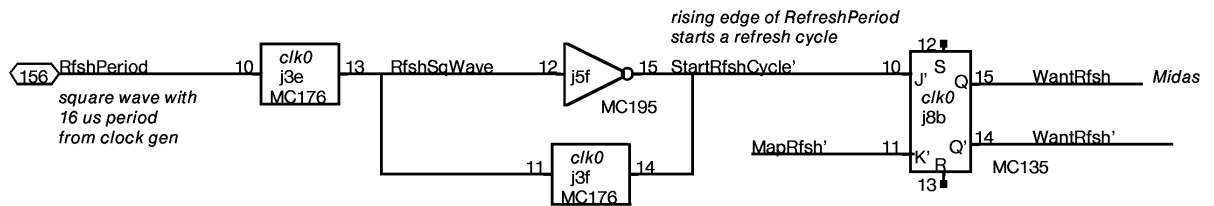
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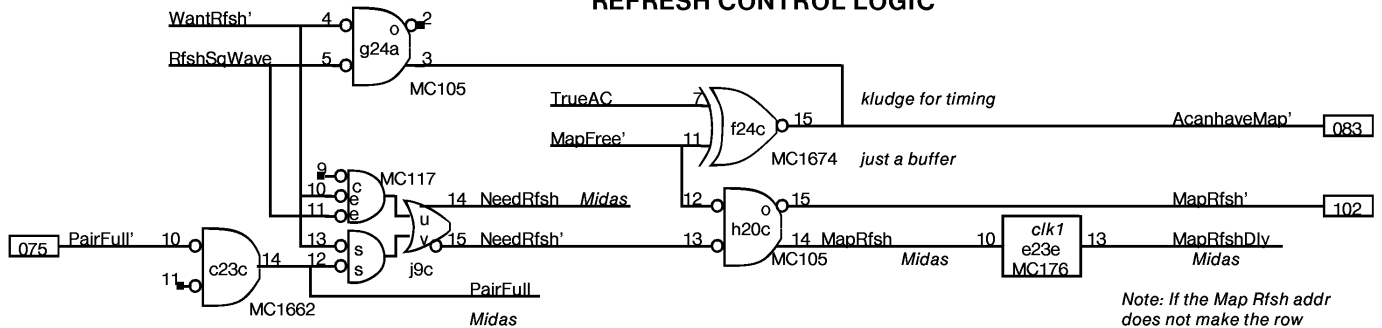






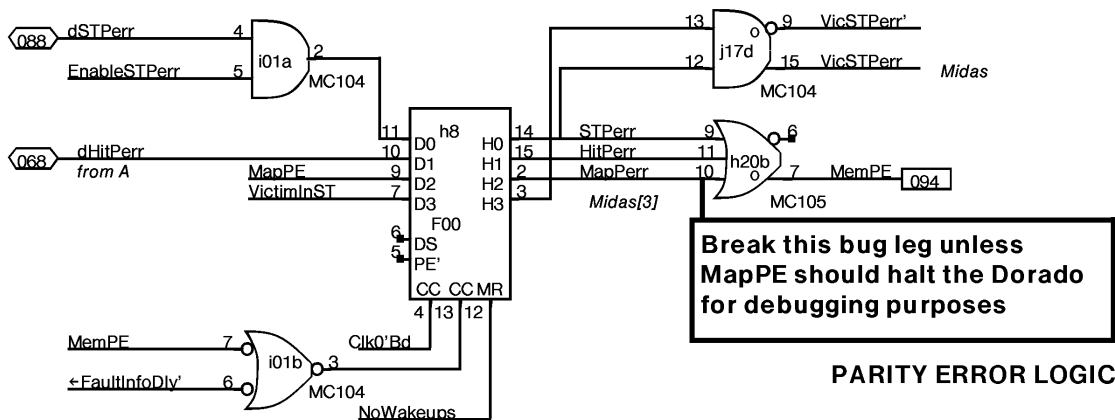
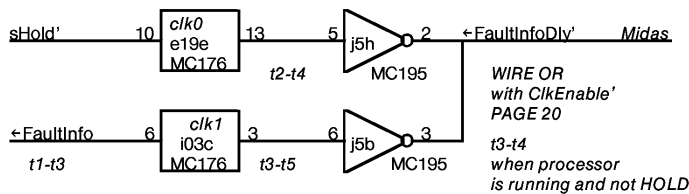


REFRESH CONTROL LOGIC

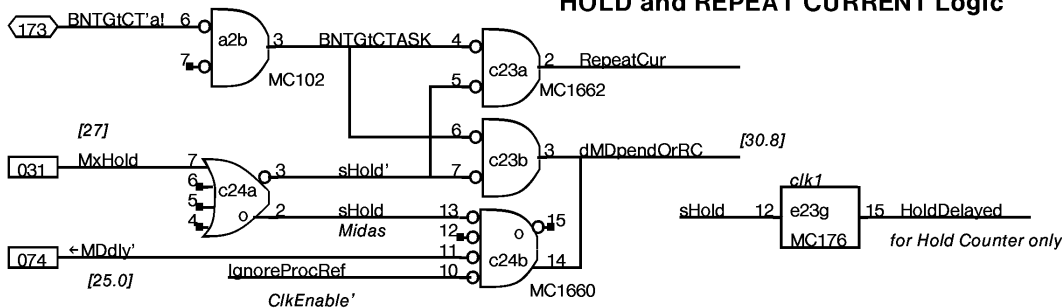


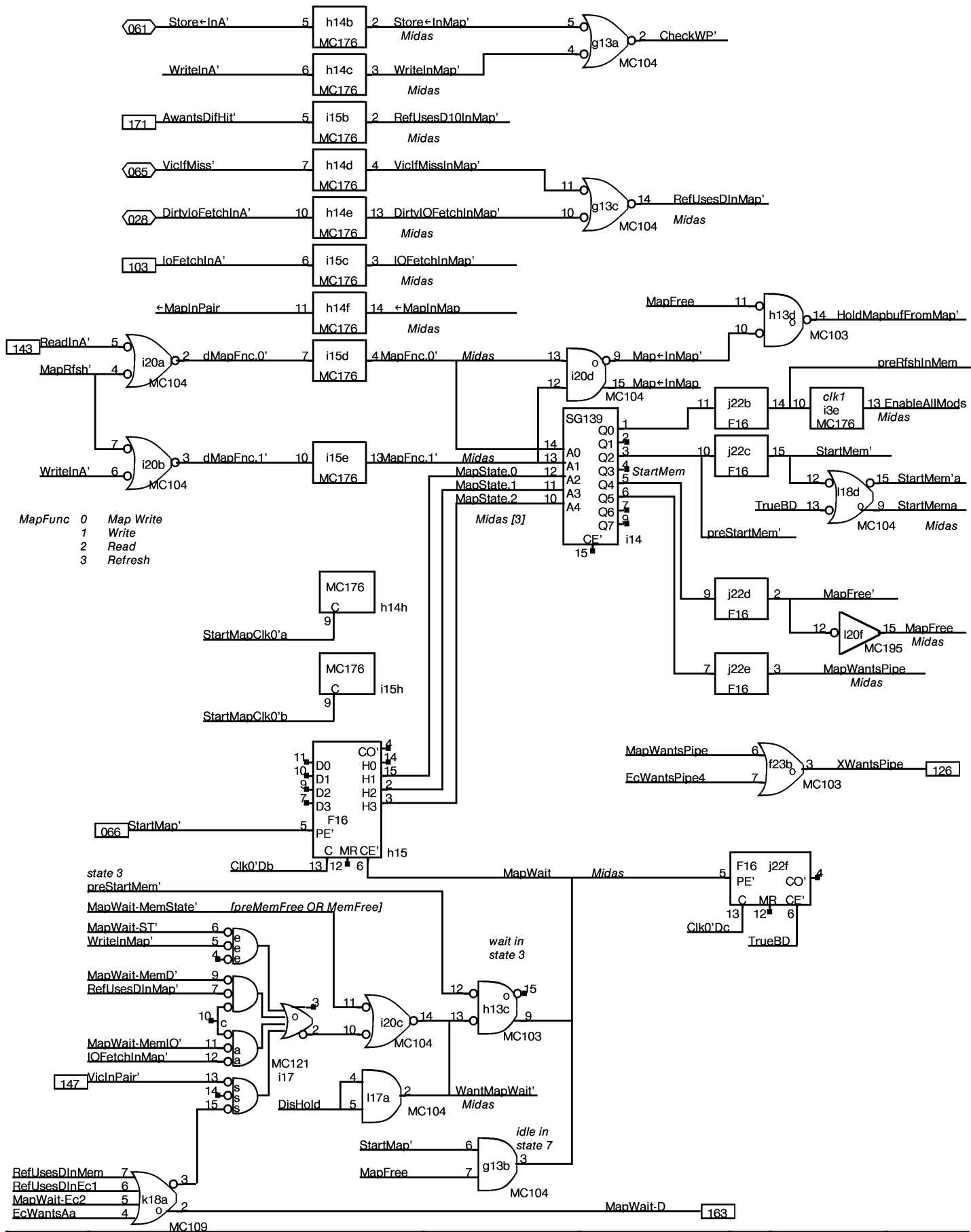
Note: If the Map Rfsh addr does not make the row address hold time, add a clk0 FF in series with this clk1 FF and OR the outputs with an OR gate.

←FAULTINFO LOGIC



HOLD and REPEAT CURRENT Logic

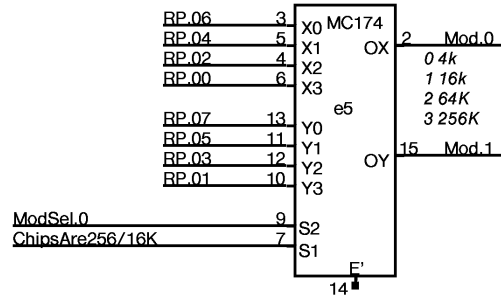




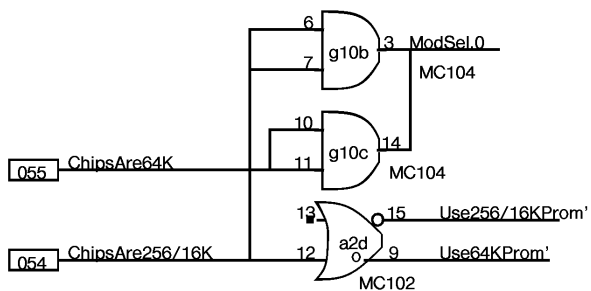


Note: Chip size options are available:
16K only, 16K or 64K, 64K only, 64K
or 256K, 256K only.

Only one of the ChipsAreXXX signals should
be true at any time.

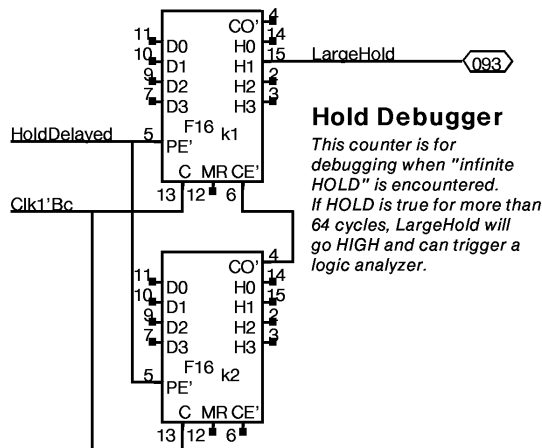


**BREAK THIS BUG LEG
UNTIL 256K CHIPS
ARE USED**



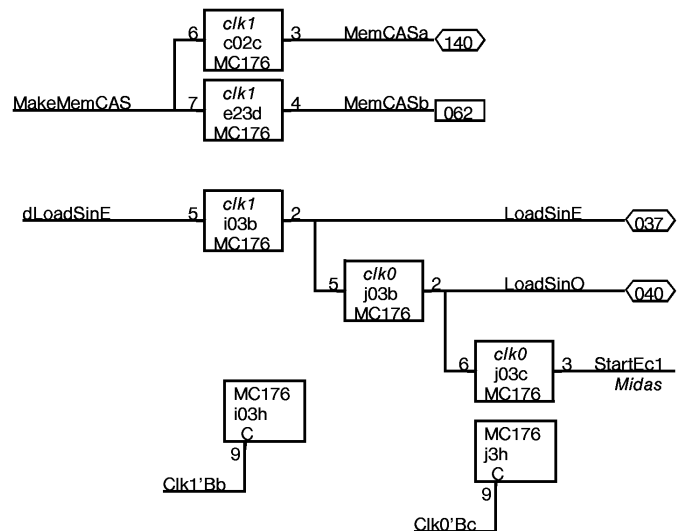
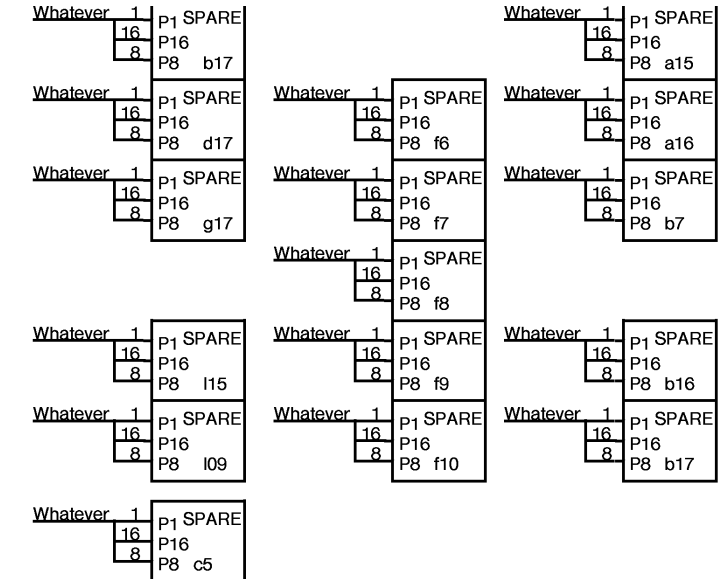
*ChipsAre256k
in the glorious future*

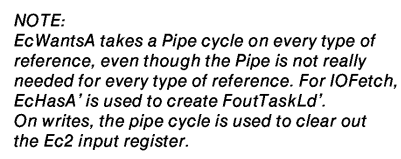
*When the only two chip
options for Dorado are
64K or 256K, then interpret
this signal as ChipsAre256K
and enable the upper gate of ModSel.0*



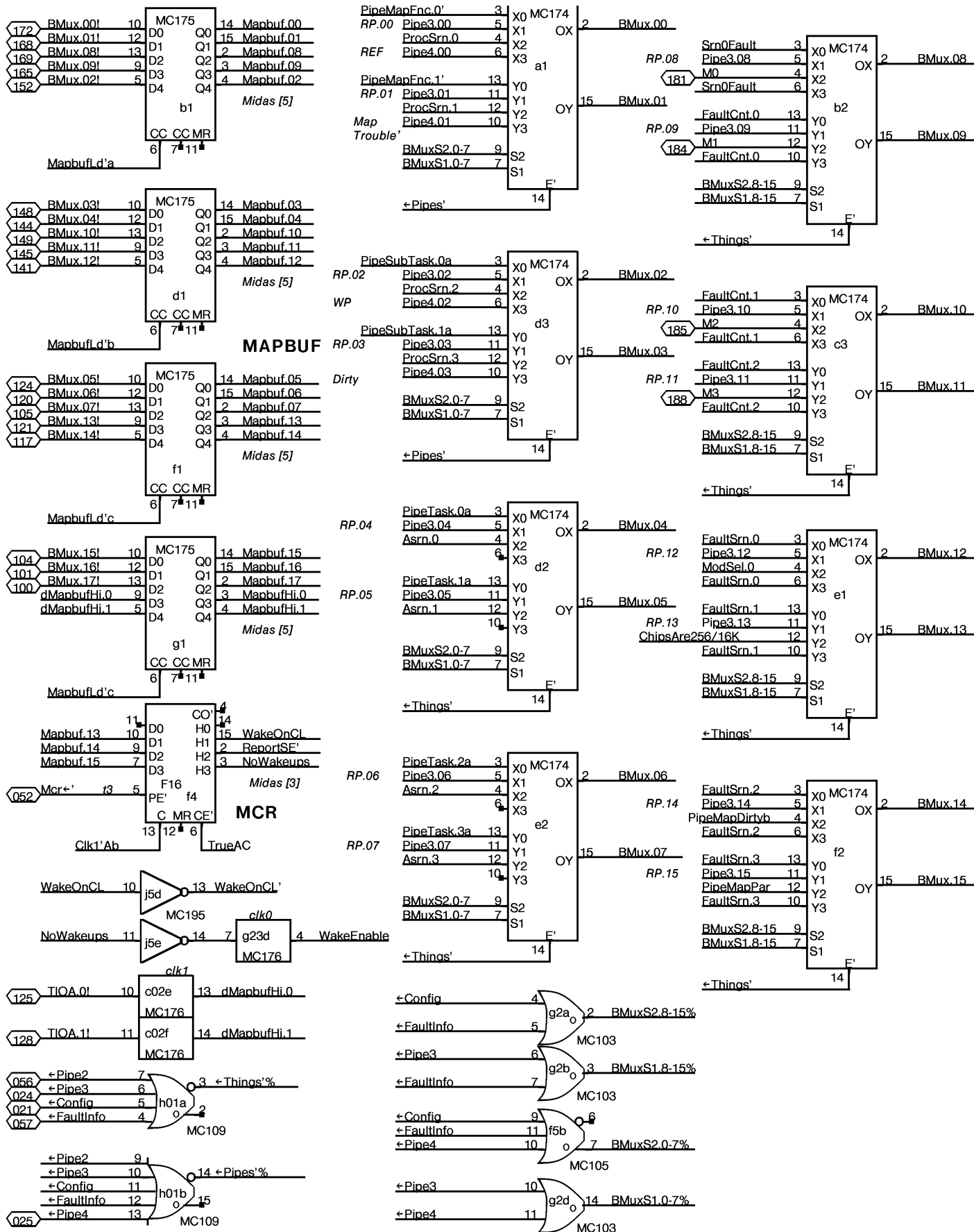
Hold Debugger

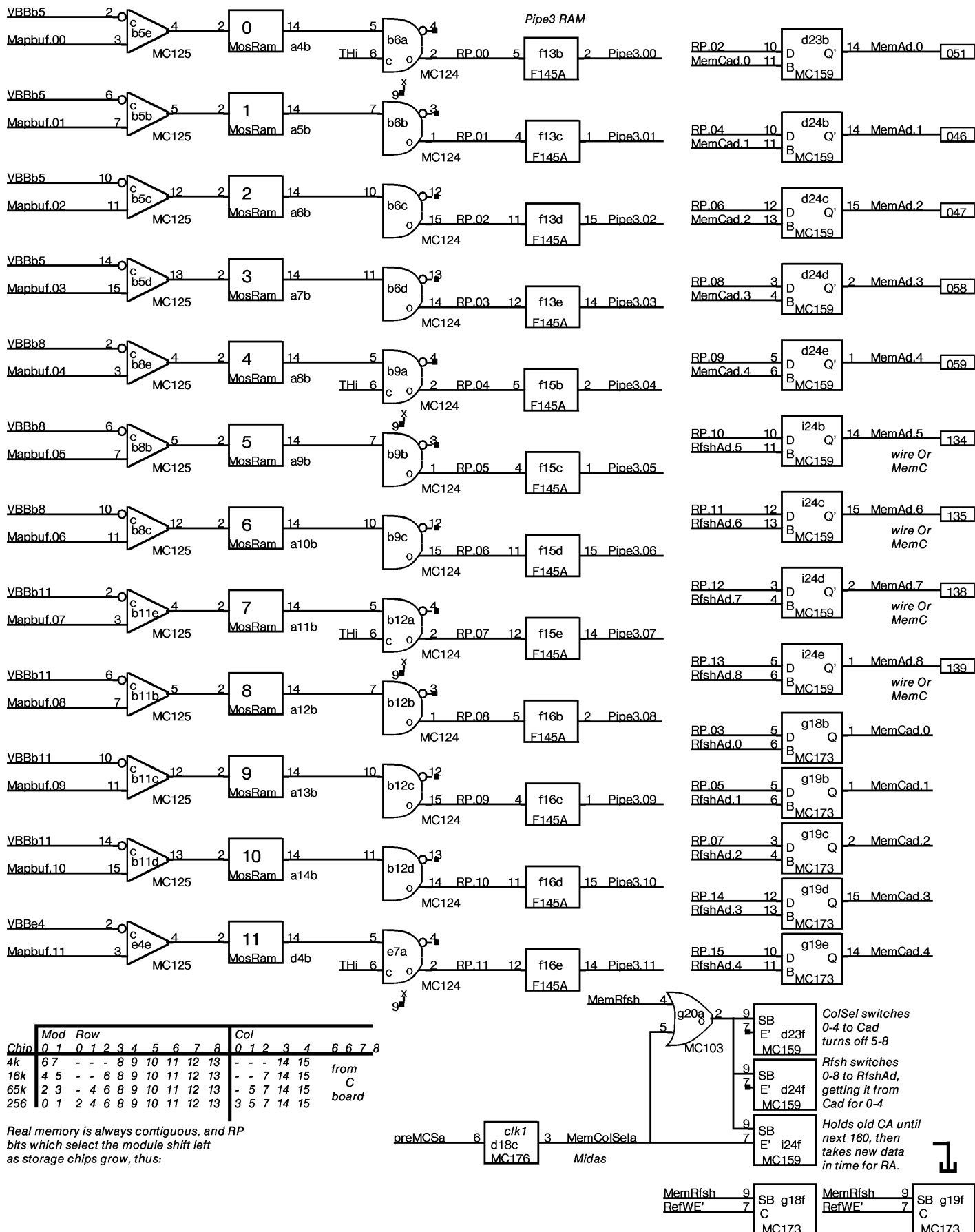
*This counter is for
debugging when "infinite
HOLD" is encountered.
If HOLD is true for more than
64 cycles, LargeHold will
go HIGH and can trigger a
logic analyzer.*

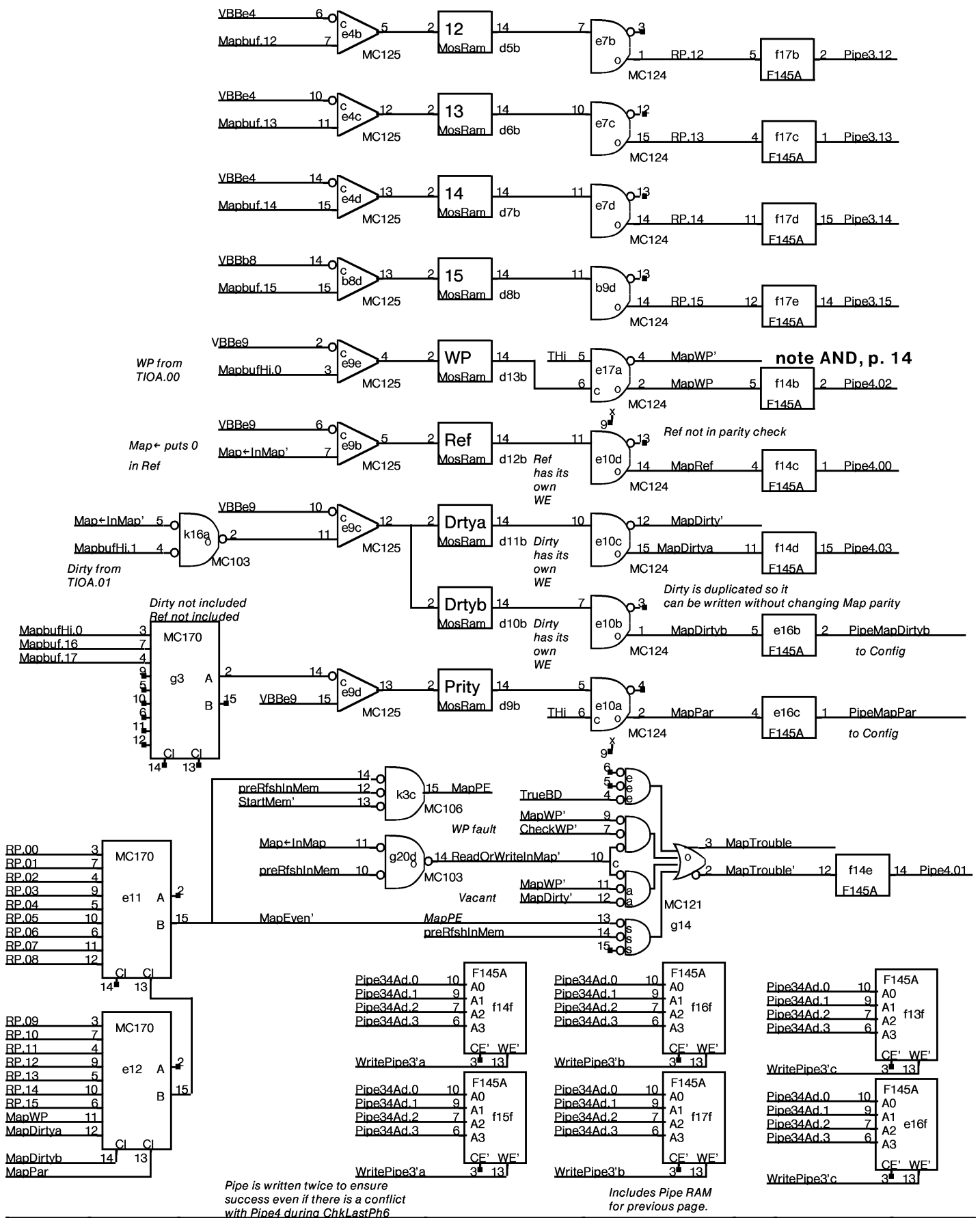


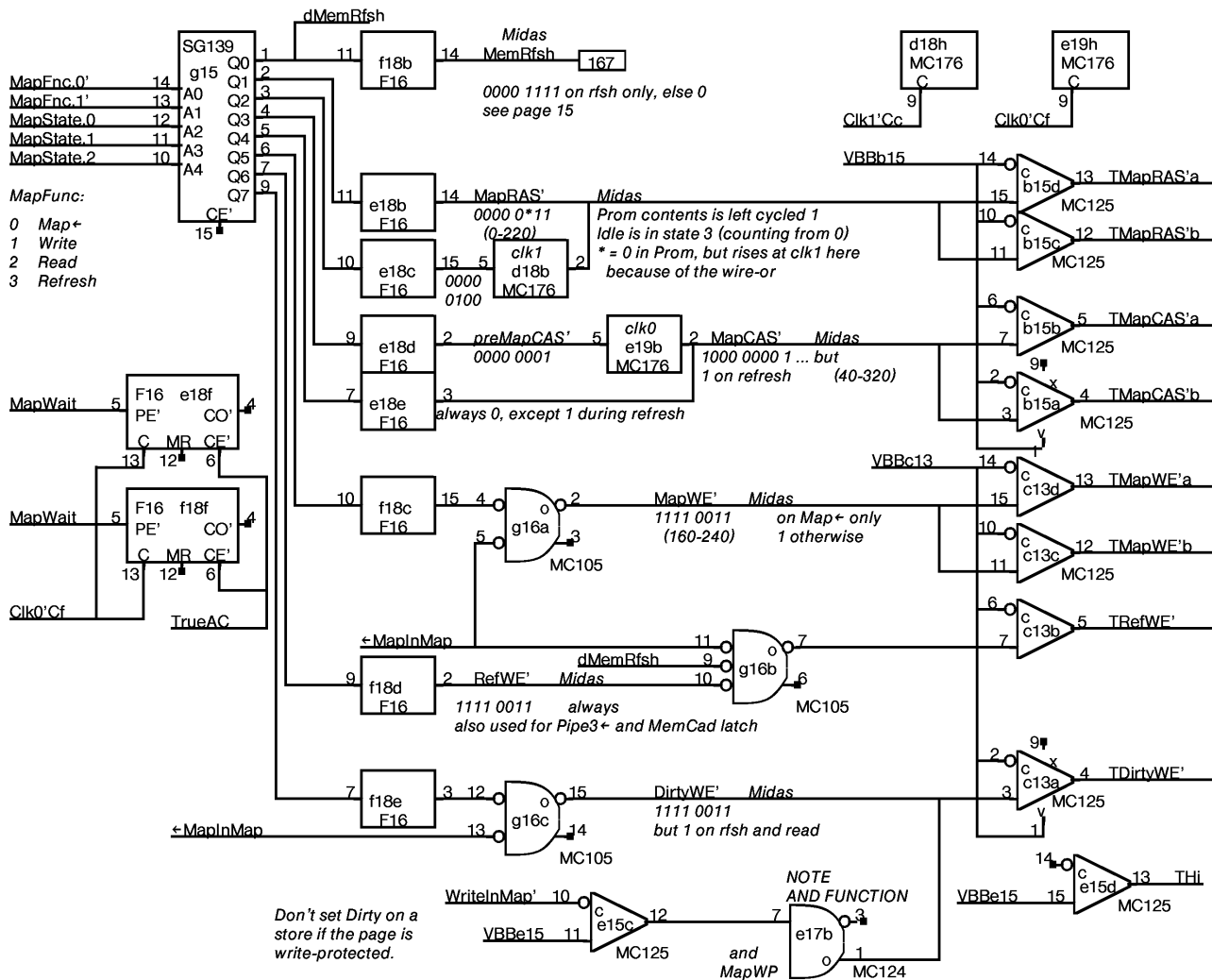




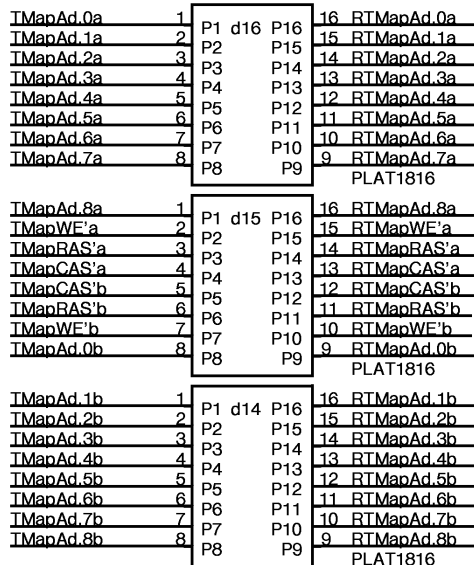




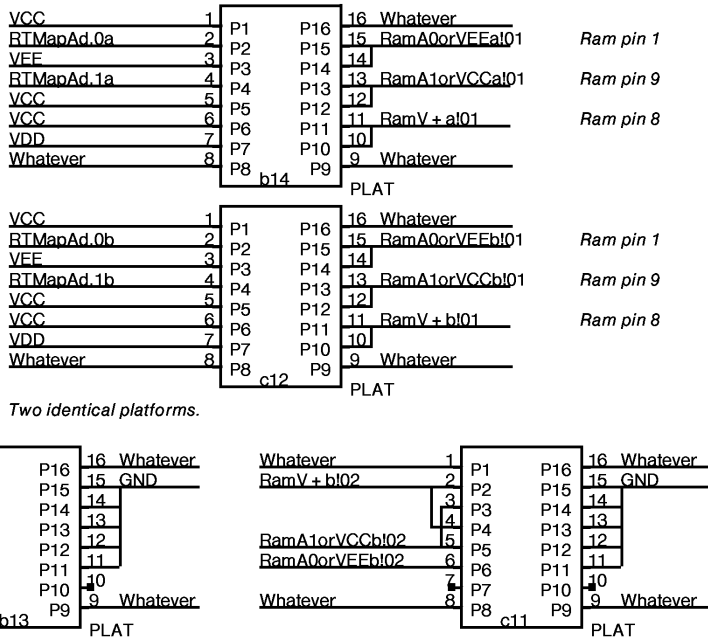


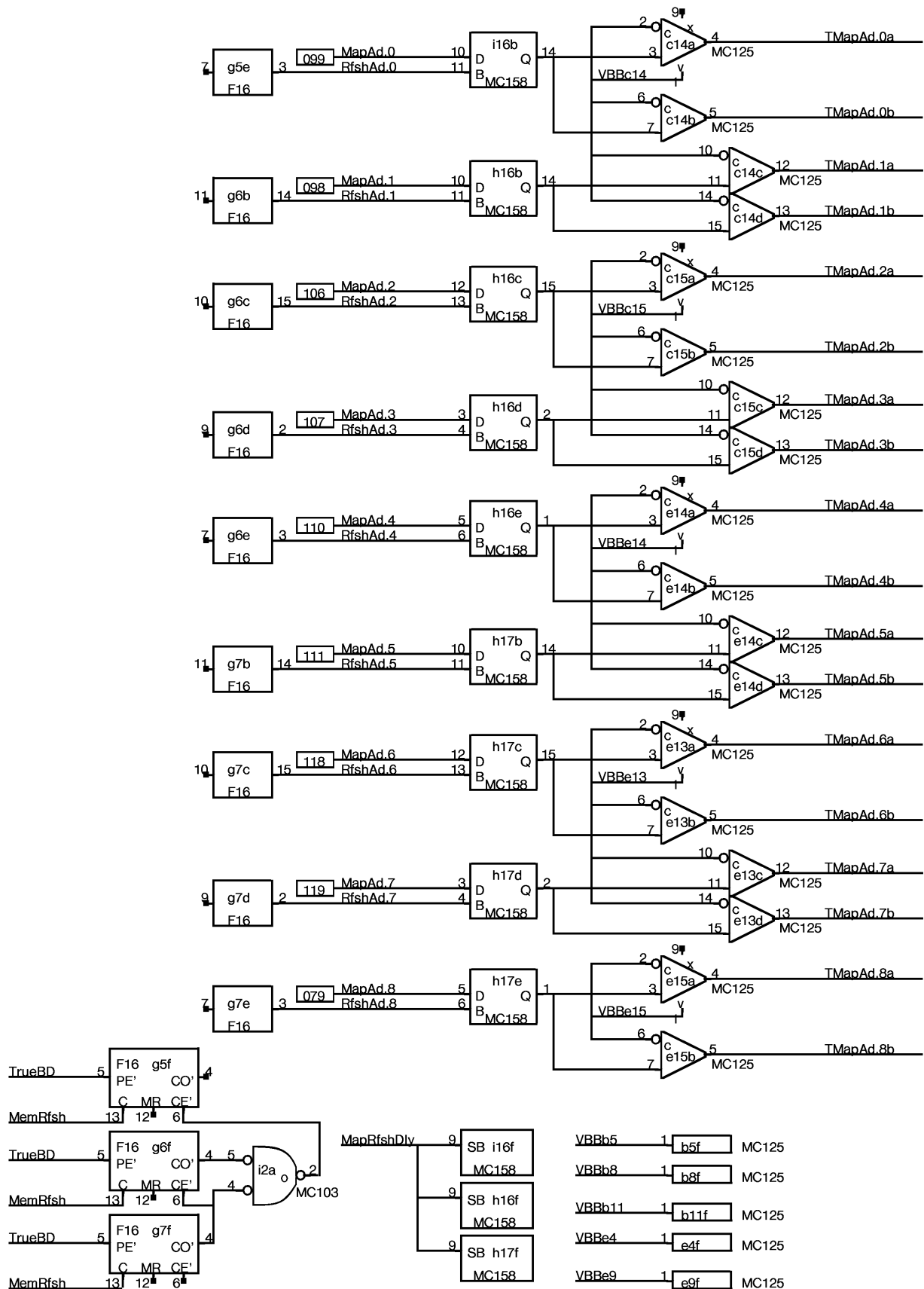


PLATS are BECKMAN 898-3-R027 series resistors 8 per chip



PLATS are Jumper wires for configuration





RamA0orVEFa15	1	A0
RamA1orVCCa14	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a118	8	V +
GND	16	GRND

RP.00

a4

RamA0orVEFa109	1	A0
RamA1orVCCa109	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a111	8	V +
GND	16	GRND

RP.05

a9

RamA0orVEFa103	1	A0
RamA1orVCCa103	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a103	8	V +
GND	16	GRND

RP.10

a14

RamA0orVEFa114	1	A0
RamA1orVCCa113	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a117	8	V +
GND	16	GRND

RP.01

a5

RamA0orVEFa108	1	A0
RamA1orVCCa107	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a109	8	V +
GND	16	GRND

RP.06

a10

GND	5	SIP	b42f
RamV + a119	8	SIP	b42i
GND	4	SIP	b42e
RamA0orVEFa116	1	SIP	b42b

GND	5	SIP	b43f
RamV + a116	8	SIP	b43i
GND	4	SIP	b43e
RamA1orVCCa115	1	SIP	b43b

RamA0orVEFa113	1	A0
RamA1orVCCa112	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a115	8	V +
GND	16	GRND

RP.02

a6

RamA0orVEFa107	1	A0
RamA1orVCCa106	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a108	8	V +
GND	16	GRND

RP.07

a11

GND	5	SIP	b44f
RamV + a113	8	SIP	b44i
GND	4	SIP	b44e
RamA0orVEFa111	1	SIP	b44b

GND	5	SIP	b45f
RamV + a110	8	SIP	b45i
GND	4	SIP	b45e
RamA1orVCCa108	1	SIP	b45b

RamA0orVEFa112	1	A0
RamA1orVCCa111	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a114	8	V +
GND	16	GRND

RP.03

a7

RamA0orVEFa105	1	A0
RamA1orVCCa106	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a106	8	V +
GND	16	GRND

RP.08

a12

GND	5	SIP	b46f
RamV + a107	8	SIP	b46i
GND	4	SIP	b46e
RamA0orVEFa106	1	SIP	b46b

GND	5	SIP	b47f
RamV + a104	8	SIP	b47i
GND	4	SIP	b47e
RamA1orVCCa104	1	SIP	b47b

RamA0orVEFa110	1	A0
RamA1orVCCa110	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a112	8	V +
GND	16	GRND

RP.04

a8

RamA0orVEFa104	1	A0
RamA1orVCCa105	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a105	8	V +
GND	16	GRND

RP.09

a13

Capacitor Options:

16K RAMs: install CAPS
between pins 5 and 8
AND between pins 1 and 4,
b42,b43,b44,b45,b46,b47

64K RAMs: install CAPS
between pins 5 and 8 ONLY,
b42,b43,b44,b45,b46,b47

256K RAMs: same as 64K.

RamA0orVEEb14	1	A0
RamA1orVCCb14	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b17	8	V +
GND	16	GRND

RP.11

d4

RamA0orVEEb108	1	A0
RamA1orVCCb109	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b110	8	V +
GND	16	GRND

Parity

d9

RamA0orVEEb113	1	A0
RamA1orVCCb113	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b116	8	V +
GND	16	GRND

RP.12

d5

RamA0orVEEb107	1	A0
RamA1orVCCb107	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
TDirtyWE'	3	WE'
RamV + b108	8	V +
GND	16	GRND

DirtyB

d10

GND	5	SIP	d42f
RamV + b118	8	SIP	d42i
GND	4	SIP	d42e
RamA0orVEEb115	1	SIP	d42b

GND	5	SIP	d43f
RamV + b115	8	SIP	d43i
GND	4	SIP	d43e
RamA1orVCCb115	1	SIP	d43b

RamA0orVEEb112	1	A0
RamA1orVCCb112	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b114	8	V +
GND	16	GRND

RP.13

d6

RamA0orVEEb106	1	A0
RamA1orVCCb106	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
TDirtyWE'	3	WE'
RamV + b107	8	V +
GND	16	GRND

DirtyA

d11

GND	5	SIP	d44f
RamV + b112	8	SIP	d44i
GND	4	SIP	d44e
RamA0orVEEb110	1	SIP	d44b

GND	5	SIP	d45f
RamV + b109	8	SIP	d45i
GND	4	SIP	d45e
RamA1orVCCb108	1	SIP	d45b

RamA0orVEEb111	1	A0
RamA1orVCCb111	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b113	8	V +
GND	16	GRND

RP.14

d7

RamA0orVEEb104	1	A0
RamA1orVCCb105	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
TRefWE'	3	WE'
RamV + b105	8	V +
GND	16	GRND

Ref

d12

GND	5	SIP	d46f
RamV + b106	8	SIP	d46i
GND	4	SIP	d46e
RamA0orVEEb105	1	SIP	d46b

GND	5	SIP	d47f
RamV + b104	8	SIP	d47i
GND	4	SIP	d47e
RamA1orVCCb104	1	SIP	d47b

RamA0orVEEb109	1	A0
RamA1orVCCb110	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b111	8	V +
GND	16	GRND

RP.15

d8

RamA0orVEEb103	1	A0
RamA1orVCCb103	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b103	8	V +
GND	16	GRND

WP

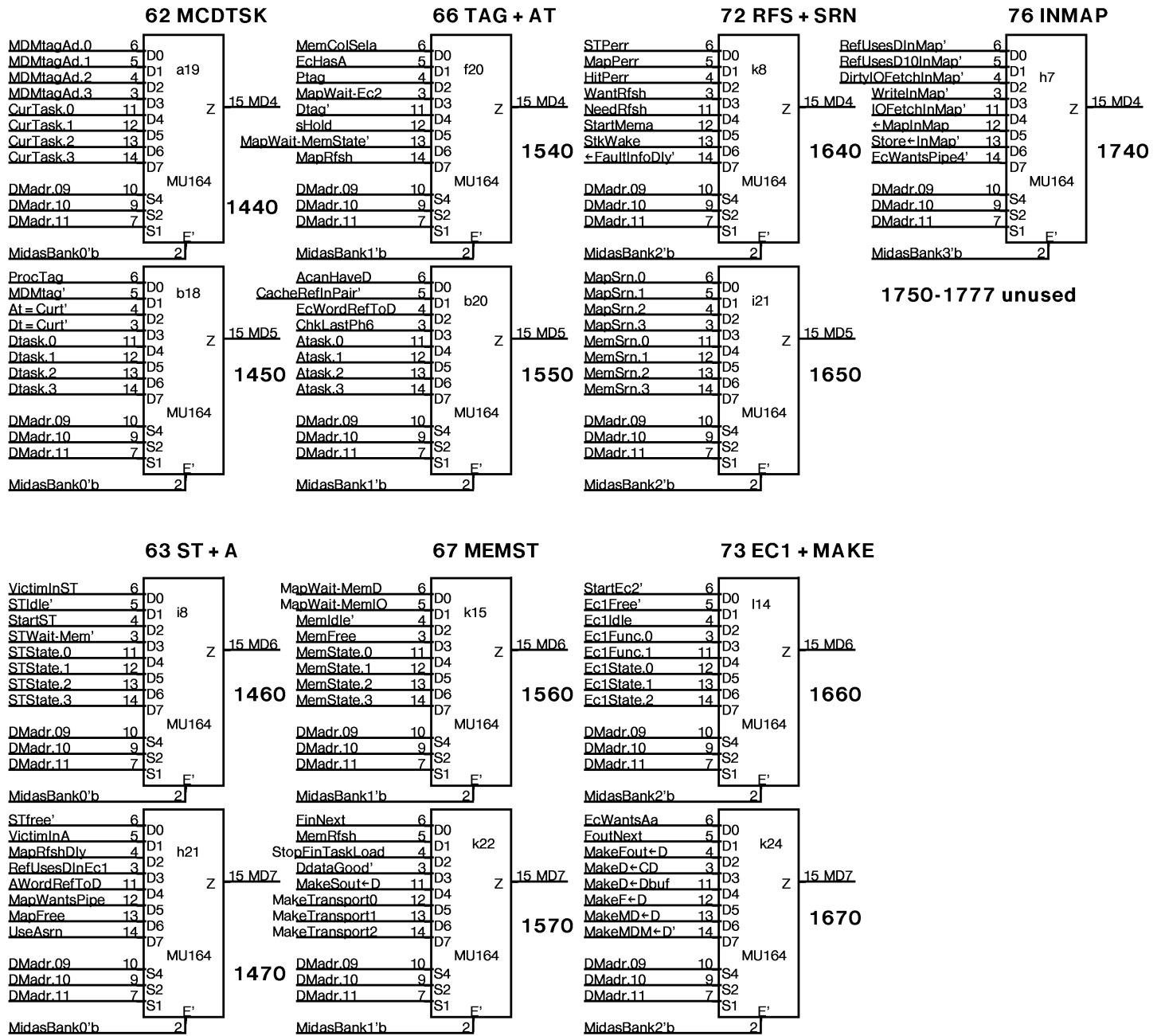
d13

Capacitor Options:

16K RAMs: install CAPS
between pins 5 and 8
AND between pins 1 and 4,
d42,d43,d44,d45,d46,d47

64K RAMs: install CAPS
between pins 5 and 8 ONLY,
d42,d43,d44,d45,d46,d47

256K RAMs: same as 64K.



Bank	Addresses
0	1400-1477
1	1500-1577
2	1600-1677
3	1700-1777

C																		C	
A	a 181	b 168	c 153	d 137	e 124	f 109	93	g	80	h	64	i	48	j	33	k	20	l	B
1	BmuxDrivr 0-1	Mapbuf 0-1-8-9	RAS-CAS A,7	Mapbuf 2-3-4-10 11-12	BmuxDrivr 12-13	Mapbuf 5-6-7-13 14-15	Mapbuf 16,17	Mapbuf 16,17	BMuxEn 11,11	109	ErrorRcvr 4,4,10,D	104	ModSinEn F16 9	HOLDcntr F16 8	ppclk/IPR 210 20				1
2	BNTGtCT a,4,c,8	BmuxDrivr 8-9	CAS-WE clk1 7,8,20 11,11,	BmuxDrivr 4-5	BmuxDrivr 6-7	BmuxDrivr 14-15	BMxSel	MU	ShiftInEven 15,9,5,5	STOut 5,5,10,10	HOLDcntr Clock								2
3	ModEn F16 7	SHIFTS LOADS 103 5,5,5,5	BmuxDrivr 10-11	BmuxDrivr 2-3	ModDcode	ModEna	ParityIn 170 13	3,3,10,20	clk1 8,4,5,6,5,, 176	8,8,10, 176 4,4,5	A,5,13 106	Ec2AuIn 176 10							3
4	0	MU Mapbuf	MU Mapbuf	11	IN	MCR 125 13	F16 11	10,10,9, 20,20,4 195	dPipe34Ad 174 3	dPipe34Ad 174 3	MU STOUT	MU MAPEC2	MU P34INEC						4
5	1	IN	+	12	Mod 174 8	A,11,C	RfshCnt F16 15	Pipe34Ad 141 3	STOut 176 5	4,7,11 11,4,4 195	FaultSrn F16 3	MidasBank 171 16							5
6	2	OUT	CLK 212 20	13		+	RfshCnt F16 15	ProcSrn 141 3	CLK 210 20	CLK 210 20	EcSrn = 0 Fsrn = 0 109 3.3	10,10,10 105							6
7	3	+	CLK 210 20	14	OUT 124 13	+	RfshCnt F16 15	MU INMAP	CLK 212 20	ReportFlt 121 3	Ec1AuIn 176 9								7
8	4	IN	125 13	15		+	MU PEEC	MemPE F00 4	MU ST + A	135 3,4	RFS + SRN 117 9	Ec1Func 117 9							8
9	5	OUT	+	PAR	IN 125 12	+	MU APESRN	Ec1Srn 141 3	Ec2Srn 141 3	LdFltCnt NeedRfsh 117 3.4	FaultCnt F16 3	+							9
10	6			DTYb	OUT 124 12	+	A,8,8,5 104	PEsrn 158 3	MemAuOut F16 7	MU STOUT	MU FLT + MEM	Ec2Func F16 10							10
11	7	IN	125 12	PLAT BYPASS	DTYa	PARITY 170 13	CLK 210 20	CLK 210 20	STProm 139 4	105	MemAuIn 176 7	7,7,9 10,10,10 195	Ec2Prom 139 10						11
12	8	OUT	124 12	PLAT SWITCH	REF	PARITY 170 13	CLK 212 20	CLK 210 20	STState F16 5	139 5	MemAuOut F16 7	MU FLT + MEM	Ec1Prom 139 9						12
13	9	PLAT BYPASS	WE 125 14	WP	ADDR 125 15	Pipe34 145 13	CheckWP 6,6,6,7 104	MapAu 7,B,6,6 103	MU MAPCTRL	MemProm 139 7	MemAuIn 176 7	MU PEEC							13
14	10	PLAT SWITCH	ADDR 125 15	RESISTOR 14	ADDR 125 15	Pipe34 145 12	MapTrbl 121 13	MapAuIn 176 6	MapProm 139 6	MemProm 139 7	clk0 7,7,,,, 176	MU EC1 + MAKE							14
15	+	RasCas 125 14	ADDR 125 15	RESISTOR 14	ADDR 15,15,14 125 14	Pipe34 145 12	MapProm 139 14	MapState F16 6	MapAuIn 176 6	STIdle MapWait 117 7.5	MU MEMST	+							15
16	+	+	MU MAPCTRL	RESISTOR 14	Pipe34 145 13	Pipe34 145 12	+Map 105 14	MapAdMux 158 15	MapAdMux 158 15	MemState F16 7	103	Ec2State F16 10							16
17	Pair 2,,, F16	+	ASRNInc 3,20 210	+	OUT 124 13	Pipe34 145 13	+	MapAdMux 158 15	MapWait 121 6	VicSTPerr 10,9,C,4 104	Ec1State F16 9	6,10,10 104 10							17
18	CurTask 141 2	MU MCDTSK	CLK 212 20	clk1 14,12,, 176	MapRCW F16 14	MapRCW F16 14	MemCad 173 12	MemSrn 141 3	CLK 210 20	CLK 210 20	VicInSt MapWaitD 109 6.9	1,1,9,6 104							18
19	MU MCDTSK	TrueNext 158 2	CLK 210 20	CLK 210 20	clk0 14,1,1,4 176 f,g	ASRN F16 3	MemCad 173 12	StartST 121 5	CLK 210 20	CLK 210 20	clk0 5*10,9 176								19
20	EMUorFT a,1,1,1 113	MU TAG + AT	CLK 210 20	Tag211 211 1,B	ProcTag F145	MU TAG + AT	12,20,C, 103 13	A,4,4 105	MapAu 104 6	Ec1AuOut 176 9	Ec1AuOut 176 9	10,9,9 9,6,1 195							20
21	Atask F16 2	Dt = Ct 113 1	PipeTask Pipe2 F145 2	PipeTaska 197 2	PipeSubT Pipe2 F145 2	MDMTtag F145 1	dMDMAd 159 1	MU ST + A	MU RFS + SRN	Ec2AuOut F16 10	Ec2AuOut F16 10	Midas 176 18							21
22	dMDMtagAd 158 1	At = Ct 113 1	FinTask F16 2	MDMtagAd 141 1	DTask F16 2	MakeMD + D 117 1	dMDMAd 100 1	MapSRN 141 3	STOut 176 5	MapAUOut F16 6	MU MEMST	Midas 164 18							22
23	ASubTask 2,2,1,1 F16	CurSubT 2,2,d,e 141	RptCur 4,4,4,1 1662	MemAd 0 159 12	CAS-WE 1,7,8,4,5,4 176 clk1	1,6,1,D 103	clk0 1,3,11,e 176 5.5	dPipe02Ad 158 3	Hit 1,1,1 105	Transport 212 5,5	5,B,1,5 102	Midas 176 18							23
24	bNext 1,1,1,1,,, 197	FinSubT 2,2,d,e F16	HOLD 1660 4	MemAd 1-4 159 12	RAS-CAS A,7 231	CacheRef 1,1,4 1674	Page1 4,1,2 105	Pipe02Ad 141 3	MemAd 5-8 159 12	FoutTask 176 2	MU EC1 + MAKE	Midas 18,18, 102 C,18							24
C																		D	
E	a 11	b 26	c 39	d 55	e 70	f 86	99	g	114	h	129	i	143	j	159	k	174	l	F

Location h11		11 cycles IO Store		12 cycles VictimStore	
ST	ADDR	0-13	14-17	20-33	34-37
Q	STATE	012345678910	11	01234567891011	
0	prepreLoadEn'	00000001110	0 1	00000000111 1	1
1	preEcLoadEn'	10101010111	1 1	11010101011 1	1
2	preShiftEn'	000000001111	1 1	100000000111 1	1
3	preEnEcGen'	11111111110	0 1	01111111111 1	1
4	preMapWait-ST'	00001111110	0 1	00000111111 0	1
5	preSTfree'	11111111101	1 1	11111111110 1	1
6	prepreSTPerrNow'	00000000111	1 1	10000000011 1	1
7	preStopFinTaskLd	11111100001	1 0	11111111000 0	0
Location i12					
0	preFinNext	11111110001	1 0	00000000000 0	0
5	preMakeSout+D	00000000000	0 0	11111111000 0	0
6	PreMakeTrnspt0	00000000000	0 0	11111111000 1	0

Location l12		Write	CacheLoad	Clean or Miss	Dirty Hit
EC1	ADDR	0 - 7	10 - 17	IOFetch	IOFetch
Q	STATE	01234567	01234567	01234567	01234567
0	preEcEn'	01111111	01111111	01111111	01111111
1	preMakeTrspt2	00000000	00111100	00000000	00111100
2	preMakeFout+D	00000000	00000000	00000000	00011100
3	preFoutNext	00000000	00000000	00011100	00001100
4	preMakeD+CD	00000000	00111100	00000000	00000000
5	preEcWantsA	01000000	01000000	01000000	01000000
6	preEc1Free'	11111101	11111101	11111101	11111101
7	preStartEc2'	11111011	11111011	11111011	11111011

Location i14		Refresh	Read	Write	Map write
MAP	ADDR	0 - 7	10 - 17	20 - 27	30 - 37
Q	STATE	01234567	01234567	01234567	01234567
0	prepreRfshInMem	00010000	00000000	00000000	00000000
1	UNUSED				
2	preStartMem'	11101111	11101111	11101111	11111111
3	preStartMem	00010000	00010000	00010000	00000000
4	preMapFree'	11111101	11111101	11111101	11111101
5	preMapWantsPipe	00000000	00110000	00110000	00110000

Location l11		Write	CacheLoad	CLEAN or Miss	Dirty Hit
EC2	ADDR	0 - 7	10 - 17	IOFetch	IOFetch
Q	STATE	01234567	01234567	01234567	01234567
0	prepreMakeTnspt1	00000000	11100001	00000000	11100001
1	preMakeFout+D	00000000	00000000	00000000	11110001
2	preFoutNext	00000000	00000000	11110001	11111001
3	preMakeD+CD	00000000	11100001	00000000	00000000
4	preEc2Free	00000010	00000010	00000010	00000010
5	preEcWantsPipe4	10000000	10000000	10000000	10000000
6	preMapWait-Ec2	00000000	11110001	00000000	11110001
7	CacheLoad'	11111111	00000010	11111111	11111111

IDENTICAL

Location g15		Refresh	Read	Write	Map Write
MAP	ADDR	0 - 7	10 - 17	20 - 27	30 - 37
Q	STATE	01234567	01234567	01234567	01234567
0	preMemRfsh	00011110	00000000	00000000	00000000
1	preRAS1'	00000110	00000110	00000110	00000110
2	preRAS2'	00001000	00001000	00001000	00001000
3	prepreCAS1'	11111110	00000010	00000010	00000010
4	prepreCAS2'	11111100	00000000	00000000	00000000
5	preMapWE'	11111111	11111111	11111111	11100111
6	preRefWE'	11100111	11100111	11100111	11100111
7	preDirtyWE'	11111111	11111111	11100111	11100111

Location j13		READ	WRITE	REFRESH
MEM	ADDR	0 - 7	0 - 7	10-17
16K chips	STATE	01234567	01234567	01234567
0	prepreMemWE'	10011111	10011111	1 1111111 1
1	preSTWait-Mem'	10111111	10111111	1 1111111 1
2	prepreMemCAS	11111100	11111100	0 00000000 0
3	prepreShiftLoadEven	00010000	00010000	0 00000000 0
4	prepreMCS	11111000	11111000	0 00000000 0
5	preMemFree	00000010	00000010	0 00000010 0
6	preMemState6'	11111011	11111011	1 11111011 1
7	preMemRAS	11110001	11110001	1 11110001 1

IDENTICAL

Location j14		READ	WRITE	REFRESH
MEM	ADDR	0 - 14	15-17	20 - 34
4K chips	STATE	0123456789	10 11 12 13-15	0123456789
0	prepreMemWE'	1100000111	1 1 1 1	1111111111 1 1 1 1
1	preSTWait-Mem'	1111101111	1 1 1 1	1111111111 1 1 1 1
2	prepreMemCAS	0001111100	0 0 0 0	0001111100 0 0 0 0
3	prepreShiftLoadEven	0000000100	0 0 0 0	0000000000 0 0 0 0
4	prepreMCS	0111111000	0 0 0 0	0111111000 0 0 0 0
5	preMemFree	0000000000	0 1 0 1	0000000000 0 1 0 1
6	prepreState7'	1111111111	1 1 1 1	1111111111 1 1 1 1
7	preMemRAS	1111111100	0 0 1 0	1111111100 0 0 1 0

IDENTICAL

XEROX	Project	Reference	File	Designer	Rev	Date	Page
PARC	Dorado	Automata program tables	Memx22.sil	K. Pier	Ch	9/08/79	22

**Dorado Memory Extension Board
Stuffing and Configuration Instructions**

1. If 256 chips are NOT installed in the Memory Storage Arrays (MSA), then
 - a. Break g10.6 (MC104) before stuffing. Label this chip as MemX-g10.
2. Break h20.10 before stuffing. Label this chip as MemX-h20.
 All SG10139 chips are PROMS which must be blown and labeled before stuffing.
 Stuff all PLAT1816 with Beckman type 898-3-R27 resistor packs or equivalent.
3. Do all the actions in one of the following three columns:

Action	16K Map Chips	64K Map Chips	256K Map Chips
Blue Wire	k4.6 to k4.16	k4.5 to k4.16	k4.4 to k4.16
Stuff into MosRam sockets	MK4116-2 or equivalent	TMS4156 or equivalent	256K Ram or equivalent
Jumper wires in socket b14	3 to 14 5 to 12 7 to 10	6 to 11 4 to 13	6 to 11 4 to 13 2 to 15
Jumper wires in socket c12	3 to 14 5 to 12 7 to 10	6 to 11 4 to 13	6 to 11 4 to 13 2 to 15
capacitors in SIP sockets b42,b43,b44,b45,b46,b47 d42,d43,d44,d45,d46,d47	0.1 mmf pins 5 to 8 each socket AND 0.1 mmf pins 1 to 4 each socket	0.1 mmf pins 5 to 8 each socket	0.1 mmf pins 5 to 8 each socket
1 KOhm resistor in socket b14 and 1 KOhm resistor in socket c12	NO	1 to 15 IF Ram has self refresh feature- Motorola or equivalent	NO
22 uf Cap in PLAT c11 & b13	+ 2 to 15 + 4 to 14 + 11 to 6	+ 3 to 14 + 5 to 12	+ 3 to 14 + 5 to 12

